

APS110 Tutorial 1

CONCEPTS OF STRESS AND STRAIN

The three principal ways in which a load may be applied:

1- tension, **2-** compression, and **3-** shear. *In engineering practice many loads are torsional rather than pure shear.

*The tensile test output is recorded as load or force versus elongation as a function of specimen size.

*To cancel out the geometrical factors, load and elongation are normalized to the respective parameters of **engineering stress** and **engineering strain** as follow:

Engineering stress:

$$\sigma = \frac{F}{A_0}$$

Engineering strain:

$$\epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0}$$

STRESS–STRAIN BEHAVIOR

* The degree to which a structure deforms depends on the **magnitude of an imposed stress**.

* The stress and strain are proportional to each other through the relationship:

>>> This is known as **Hooke's law**, and the constant of proportionality ***E* (GPa or psi)** is the **modulus of elasticity, or Young's modulus**.

ELASTIC PROPERTIES OF MATERIALS

* This process of gently applying force to an object that can go back to its original shape (**reversible**) is called **elastic behavior**.

Types of Materials (you can mention this very quickly, just to show what's to come)

- **Metals:**

- Strong, ductile
- high thermal & electrical conductivity
- opaque, reflective.

- **Polymers/plastics:** Covalent bonding sharing of e's

- Soft, ductile, low strength, low density
- thermal & electrical insulators
- Optically translucent or transparent.

- **Ceramics:** ionic bonding (refractory) – compounds of metallic & nonmetallic elements (oxides, carbides, nitrides, sulfides)

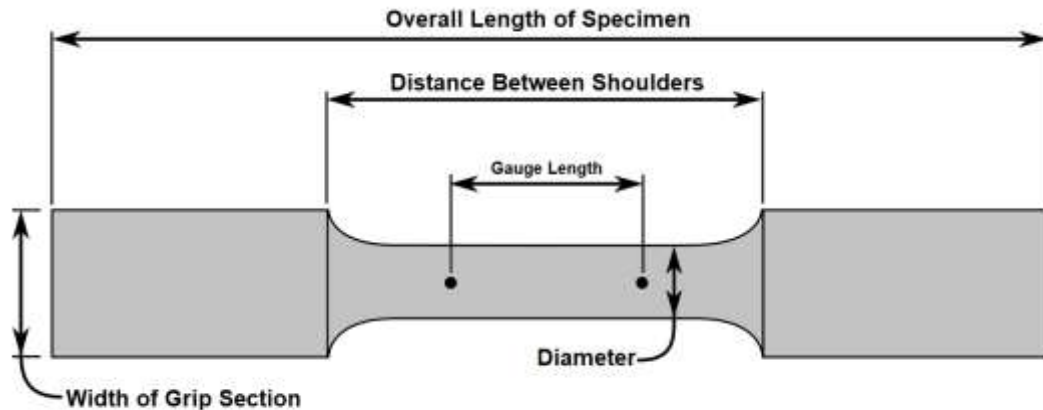
- Brittle, glassy, elastic
- non-conducting (insulators)

Learning objectives of the week (accessible in the syllabus section in Quercus)

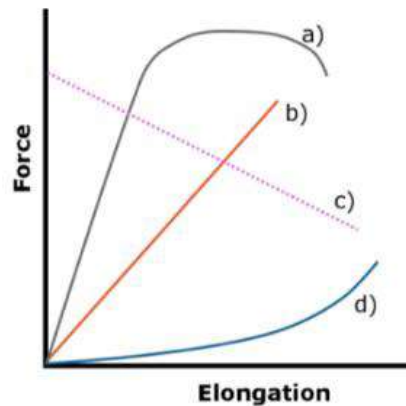
Make sure you're familiar with these concepts for your tutorials

- Define Hooke's law, engineering stress, and engineering strain - Explain how elastic stress and strain are related through Hooke's law
- Explain how stress, and strain are sample size independent, while force and displacement are not (mention this in tutorial 1 for sure)
- Describe the conventional uniaxial tensile test - Identify and explain the following on a typical "dog-bone" tensile specimen: reduced section, gauge length, cross sectional area, grip region
- Describe elastic deformation in terms of macroscopic behaviour as well as in terms of atomic positions - Sketch a generalized interatomic force-vs.-separation curve for two atoms in a solid - Using the interatomic force-separation curve, describe how the Young's Modulus is a structure independent property
- Predict, given Young's modulus, the strain resulting from a given applied elastic stress, or vice-versa - Differentiate between strength, stiffness, strain, and Young's modulus - Calculate engineering stress and engineering strain - Illustrate how Young's modulus is a structure independent property

1. On a standard metallic tensile sample, the reduced section corresponds to which of the following?
- a) The region between the grip regions where the cross-sectional area is reduced
 - b) The region where the cross-sectional area is reduced through localized plastic deformation
 - c) The region between the ends of a strain gauge
 - d) All of the above



2. If you were to hang four weights of different mass on a spring and plot force applied versus the elongation of the spring, which of the following curves would you expect to see? a) and b) depending on the weights and spring



3. You are given a sample of a metal bar with dimensions of $w: 5 \text{ mm} \times h: 4 \text{ mm} \times l: 20 \text{ cm}$, and you need to test whether or not the material can withstand a stress of 80 MPa . Assuming you apply a load along the long axis, what load would you need to apply to achieve this stress?
- a) 1600 N
 - b) 1 600 000 N
 - c) 4 000 000 N
 - d) 4000 N

4. Material X has a higher Young's modulus than material Y. Which of the following statements is necessarily TRUE?

a) Both materials are loaded in tension beyond the yield strength, but without necking. Upon unloading, the Young's modulus of both materials decreases to some extent.

b) Material X resists plastic deformation more than material Y.

c) Both a) & b).

d) If we prepare two specimens with similar length and diameter, material X will experience lower strain than material Y, if we stay in the elastic region

5. On a conventional metallic tensile specimen, the reason for the reduced section is best described by which of the following?

a) The reduced cross-sectional area allows careful study of a region where we can expect the majority of the strain to occur.

b) The reduced section corresponds to a section where the stress is reduced, allowing careful study of a region where we can expect the majority of the strain to occur.

c) The reduced cross-sectional area allows careful study of a region where the force in the sample will be highest.

d) The reduced section corresponds to a section where the stress is reduced, allowing careful study of a region where the force in the sample will be highest.

6. Which of the following could be considered a scientific model?

a) Both b) and c)

b) Atoms as hard spheres

c) Net interatomic force as a mechanical spring

d) Neither b) nor c)

7. Which of the following is the best atomic level definition of Young's modulus?

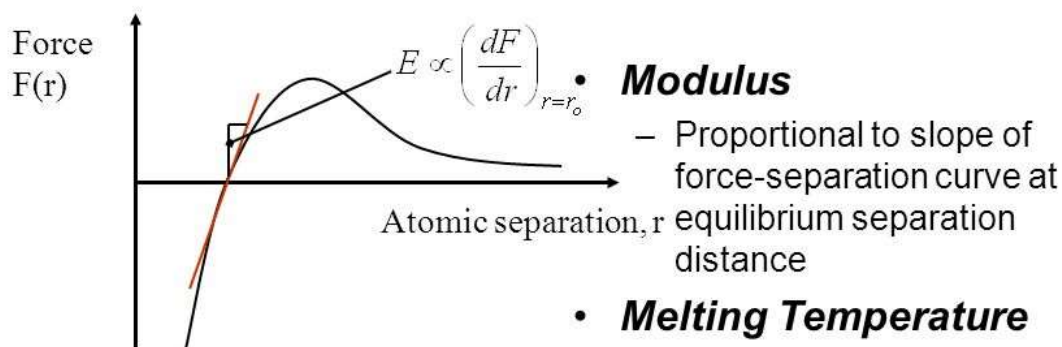
a) Inversely proportional to the slope of the interatomic force-separation curve at $r=r_0$.

b) Directly proportional to the slope of the interatomic force-separation curve at $r=r_0$.

c) Inversely proportional to the slope of the interatomic force-separation curve at $r=0$.

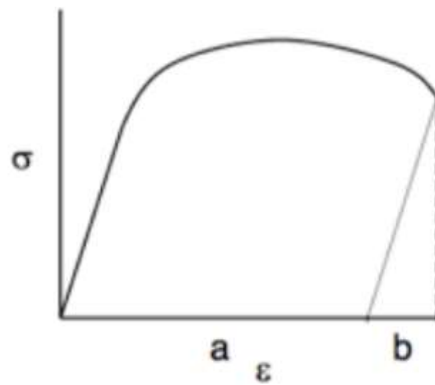
d) Directly proportional to the slope of the interatomic force-separation curve at $r=0$.

Atomistic Origins of Properties



8. Identify the strain quantity labelled “b” in the figure below.

- a) Plastic strain
- b) Elastic strain
- c) Young-Abeille’s strain
- d) Total strain



Longer questions

A. Ask example 6.1 with the table below. They should try to calculate the stress and strain using those units and then plot the graph. At the end of the exercise you can show the plot from part B as the right answer.

Example 6-1 Tensile Testing of Aluminum Alloy

Convert the load and change in length data in Table 6-1 to engineering stress and strain and plot a stress strain curve.

SOLUTION

For the 1000-lb load:

$$S = \frac{F}{A_0} = \frac{1000 \text{ lb}}{(\pi/4)(0.505 \text{ in.})^2} = 4,993 \text{ psi}$$
$$e = \frac{\Delta l}{l_0} = \frac{0.001 \text{ in.}}{2.000 \text{ in.}} = 0.0005 \text{ in./in.}$$

The results of similar calculations for each of the remaining loads are given in Table 6-1 and are plotted in Figure 6-5.

Table 6-1 The results of a tensile test of a 0.505-in. diameter aluminum alloy test bar, initial length (l_0) = 2 in.

Load (lb)	Change in Length (in.)	Calculated	
		Stress (psi)	Strain (in./in.)
0	0.000	0	0
1000	0.001	4,993	0.0005
3000	0.003	14,978	0.0015
5000	0.005	24,963	0.0025
7000	0.007	34,948	0.0035
7500	0.030	37,445	0.0150
7900	0.080	39,442	0.0400
8000 (maximum load)	0.120	39,941	0.0600
7950	0.160	39,691	0.0800
7600 (fracture)	0.205	37,944	0.1025

B Do example 6-2

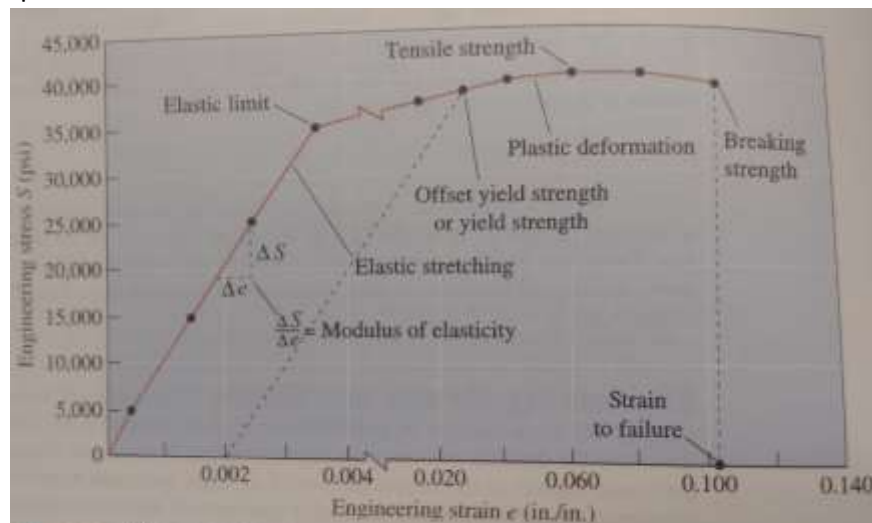


Figure 6-5 The engineering stress strain curve for an aluminum alloy from Table 6-1.

Example 6-2 Design of a Suspension Rod

An aluminum rod is to withstand an applied force of 45,000 pounds. The engineering stress strain curve for the aluminum alloy to be used is shown in Figure 6-5. To ensure safety, the maximum allowable stress on the rod is limited to 25,000 psi, which is below the yield strength of the aluminum. The rod must be at least 150 in. long but must deform elastically no more than 0.25 in. when the force is applied. Design an appropriate rod.

SOLUTION

From the definition of engineering strain,

$$e = \frac{\Delta l}{l_0}$$

For a rod that is 150 in. long, the strain that corresponds to an extension of 0.25 in. is

$$e = \frac{0.25 \text{ in.}}{150 \text{ in.}} = 0.00167$$

According to Figure 6-5, this strain is purely elastic, and the corresponding stress value is approximately 17,000 psi, which is below the 25,000 psi limit. We use the definition of engineering stress to calculate the required cross-sectional area of the rod:

$$S = \frac{F}{A_0}$$

Note that the stress must not exceed 17,000 psi, or consequently, the deflection will be greater than 0.25 in. Rearranging,

$$A_0 = \frac{F}{S} = \frac{45,000 \text{ lb}}{17,000 \text{ psi}} = 2.65 \text{ in.}^2$$

The rod can be produced in various shapes, provided that the cross-sectional area is 2.65 in.². For a round cross section, the minimum diameter to ensure that the stress is not too high is

$$A_0 = \frac{\pi d^2}{4} = 2.65 \text{ in.}^2 \quad \text{or} \quad d = 1.84 \text{ in.}$$

Thus, one possible design that meets all of the specified criteria is a suspension rod that is 150 in. long with a diameter of 1.84 in.

20. The following figure illustrates the stress-strain behaviour for a hypothetical metallic specimen that has a diameter of 1 cm and an initial (unloaded) length of 10 cm.

- Calculate the length of the specimen an instant prior to fracture (while the load is still applied), in cm.
- What is the maximum force, in kN, that this specimen can support before necking begins?

